

Statistics II: Introduction to Inference

Problem set 4

1. Let $X_1, \dots, X_n$ be an i.i.d. sample from the following distributions. In each case, find the method of moments estimator (MOME) for $g(\theta)$:

- (a) **Gamma** (α, β) , and $g(\theta) = (\alpha, \beta)^\top$.
- (b) **Beta** (α, β) and $g(\theta) = \alpha/\beta$.
- (c) **Poisson** (λ) and $g(\theta) = \exp\{-\lambda\}$.
- (d) **Location-scale Exponential** (μ, σ) with p.d.f.

$$f_{\mathbf{X}}(x; \mu, \sigma) = \begin{cases} \sigma^{-1} \exp\{-\sigma^{-1}(x - \mu)\} & x > \mu, \\ 0 & \text{otherwise,} \end{cases}$$

and $g(\theta) = (\mu, \sigma)$.

2. Let $X_1, \dots, X_n$ be an i.i.d. sample from the following distributions. In each case, find the MLE for $g(\theta)$:

- (a) **Binomial** (m, θ) , and $g(\theta) = \theta$.
- (b) **Binomial** (θ, p) , and $g(\theta) = \theta$ when $n = 1$.
- (c) **Binomial** (m, θ) , and $g(\theta) = P(X_1 + X_2 = 0)$.
- (d) **Hypergeometric** (m, r, θ) with p.m.f.

$$f_X(x; m, r, \theta) = \frac{\binom{m}{x} \binom{\theta-m}{r-x}}{\binom{\theta}{r}}, \quad \theta = m+1, m+2, \dots; \quad \max\{0, r+m-\theta\} \leq x \leq \min\{m, r\},$$

$g(\theta) = \theta$ and $n = 1$.

- (e) Double exponential: pdf $f_X(x; \theta) = 2^{-1} \exp\{-|x - \theta|\}$; with $x \in \mathbb{R}$ and $\theta \in \mathbb{R}$.
 - (f) **Uniform** (α, β) , and $g(\theta) = \alpha + \beta$.
 - (g) **Normal** (θ, θ^2) , and $g(\theta) = \theta$.
 - (h) **Inverse Gaussian** (θ_1, θ_2) and $g(\theta) = (\theta_1, \theta_2)$.
 - (i) **Uniform** $(-\theta, \theta)$ where $g(\theta) = \theta$.
 - (j) **Uniform** $(\theta, \theta + 1)$ where $g(\theta) = \theta$. Show that the median is not unique here.
3. Let $\begin{pmatrix} X_1 \\ Y_1 \end{pmatrix}, \dots, \begin{pmatrix} X_n \\ Y_n \end{pmatrix}$ be a random sample of size n from **multinomial** (m, p_1, p_2) distribution, with common p.m.f.

$$f_{(X,Y)}(x, y) = \frac{m!}{x!y!(m-x-y)!} p_1^x p_2^y (1-p_1-p_2)^{m-x-y}, \quad x, y \in \{0, 1, \dots, m\}, \quad x+y \leq m,$$

$$0 < p_1, p_2 < 1, \quad p_1 + p_2 < 1.$$

Find the MLE of $\theta = (p_1, p_2)'$.

4. Suppose that the random variables $Y_1, \dots, Y_n$ satisfy $Y_i = \beta x_i + \epsilon_i$, $i = 1, \dots, n$ where $x_1, \dots, x_n$ are fixed constants, and $\epsilon_1, \dots, \epsilon_n$ are iid $N(0, \sigma^2)$, σ^2 unknown.
 - (a) Find a two-dimensional sufficient statistic for (β, σ^2) .
 - (b) Find the MLE of β , and show that it is an unbiased estimator of β .
 - (c) Find the distribution of the MLE of β .
 - (d) Show that $\sum Y_i / \sum x_i$ is an unbiased estimator of β .
 - (e) Calculate the exact variance of $\sum Y_i / \sum x_i$ and compare it to the variance of the MLE.
 - (f) Show that $[\sum (Y_i/x_i)]/n$ is also an unbiased estimator of β .
 - (g) Calculate the exact variance of $[\sum (Y_i/x_i)]/n$ and compare it to the variances of the estimators in the previous two estimates.

5. For a random variable X , the following are known

$$P(X \geq 0) = 1 \quad \text{and} \quad P(X \geq 10) = 1/5.$$

Prove that $E(X) \geq 2$.

6. Suppose that X is a random variable for which $E(X) = 10$, $P(X \leq 7) = 0.2$ and $P(X \geq 13) = 0.3$. Then show that $\text{var}(X) \geq 9/2$.
7. Consider a probability distribution with CDF F , expectation μ and variance σ^2 . Find the minimum size, say n , of random samples which ensures at least 0.99 probability of the event that the sample mean $\bar{X}_n$ will lie within 2σ limit of the expectation μ , i.e., $\mu - 2\sigma \leq \bar{X}_n \leq \mu + 2\sigma$.
8. Suppose that 75% of the people in a certain metropolitan area live in the city and 25% of the people live in the suburbs. If 1200 people attending a certain concert represent a random sample from the metropolitan area, what is the probability that the number of people from the suburbs attending the concert will be fewer than 270?
9. Suppose that a random sample of size n is to be taken from a distribution with mean is μ and the standard deviation is $\sigma = 3$. Use the central limit theorem to determine approximately the smallest value of n for which the following relation will be satisfied:

$$P(|\bar{X}_n - \mu| < 0.3) \geq 0.95.$$

10. Suppose that the proportion of defective items in a large manufactured lot is 0.1. What is the smallest random sample of items that must be taken from the lot in order for the probability to be at least 0.99 that the proportion of defective items in the sample will be less than 0.13?
11. Consider the random variable with X with the following specifications $E(X) = \mu$ and the r -th order central moments μ_r are $\mu_2 = 5/4$, $\mu_4 = 125/2$. What is the best possible upperbound of the probability of the event that $\bar{X}_n \in [\mu - 1, \mu + 1]$ which can be obtained using a random sample of size $n = 20$.

★ These problems are not included in midsem.

★ These problems are newly added.